Sales & Tax ID Lookup & Data Cleaning Solutions

Role: Senior Business Intelligence & Excel Specialist | Technical Specification & Reference

BI ENTERPRISE SPEC

Reconciliation Context & Business Rules:

- **Sources:** `orders_raw.csv` (Customer ID in Col B) and `tax_ids_reference.csv` (Customer ID in Col A, Tax ID in Col C).
- **Data Quality Challenge:** Irregular leading/trailing spaces (e.g. `"CUST001 "`, `" CUST003"`, `"CUST004 "`) break exact lookups and cause `#N/A` errors.
- **Requirement:** Missing/unregistered IDs (e.g. `"CUST999"`) must output **"Pending Verification"** rather than error codes.

1. Modern Excel Formula (Cell orders!E2)

```
=XLOOKUP(TRIM(B2), tax_ids_reference!$A:$A, tax_ids_reference!$C:$C, "Pending Verification")
```

- **`TRIM(B2)`:** Sanitizes Customer ID by removing leading, trailing, and duplicate ASCII spaces.
- **`tax_ids_reference!$A:$A`:** Lookup range in reference sheet (Column A).
- **`tax_ids_reference!$C:$C`:** Return range supplying corresponding Tax ID (Column C).
- **`"Pending Verification"`:** Built-in 4th parameter (`[if_not_found]`) that cleanly handles missing keys.
- **Exact Match Default:** Operates in exact match mode by default (`match_mode = 0`), guaranteeing precision.
- **Table Syntax Alternative:** `=XLOOKUP(TRIM([@[Customer ID]]), TaxTable[Customer ID], TaxTable[Tax ID], "Pending Verification")`

2. Legacy Excel Formula (Excel 2019, 2016, 2013 & Older)

```
=IFERROR(INDEX(tax_ids_reference!$C:$C, MATCH(TRIM(B2), tax_ids_reference!$A:$A, 0)), "Pending Verification")
```

- **`TRIM(B2)`:** Normalizes the lookup key in cell B2.
- **`MATCH(..., tax_ids_reference!$A:$A, 0)`:** Performs an exact search (mode `0`) in Col A and returns relative row index.
- **`INDEX(tax_ids_reference!$C:$C, ...)`:** Extracts the Tax ID at the matched row index from Col C.
- **`IFERROR(..., "Pending Verification")`:** Traps `#N/A` errors caused by unregistered customers and outputs fallback text.

3. Automated 5-Line Python pandas Pipeline

```
import pandas as pd

orders, tax = pd.read_csv('orders_raw.csv'), pd.read_csv('tax_ids_reference.csv')
orders['Customer ID'] = orders['Customer ID'].astype(str).str.strip()
merged = orders.merge(tax[['Customer ID', 'Tax ID']], on='Customer ID', how='left').fillna({'Tax ID': 'Pending Verification'})
merged.to_csv('orders_enriched.csv', index=False)
```

- **Line 1:** Imports the pandas library.
- **Line 2:** Concurrently ingests both source CSV files into memory.
- **Line 3:** Vectorized `.str.strip()` eliminates all irregular whitespace across Column B.
- **Line 4:** Executes a left join on `Customer ID`, projecting `Tax ID` and imputing `"Pending Verification"` for missing IDs.
- **Line 5:** Persists the reconciled dataset directly to `orders_enriched.csv`.

Senior BI Specialist Pro-Tip — Handling Non-Breaking Spaces (`CHAR(160)`):
Data extracted from ERP systems or web HTML tables often contains ASCII 160 non-breaking spaces (` `), which standard `TRIM()` ignores.

- **Excel Formula:** `TRIM(SUBSTITUTE(B2, CHAR(160), " "))`
- **Python pandas:** `orders['Customer ID'] = orders['Customer ID'].astype(str).str.replace('\xa0', ' ').str.strip()`